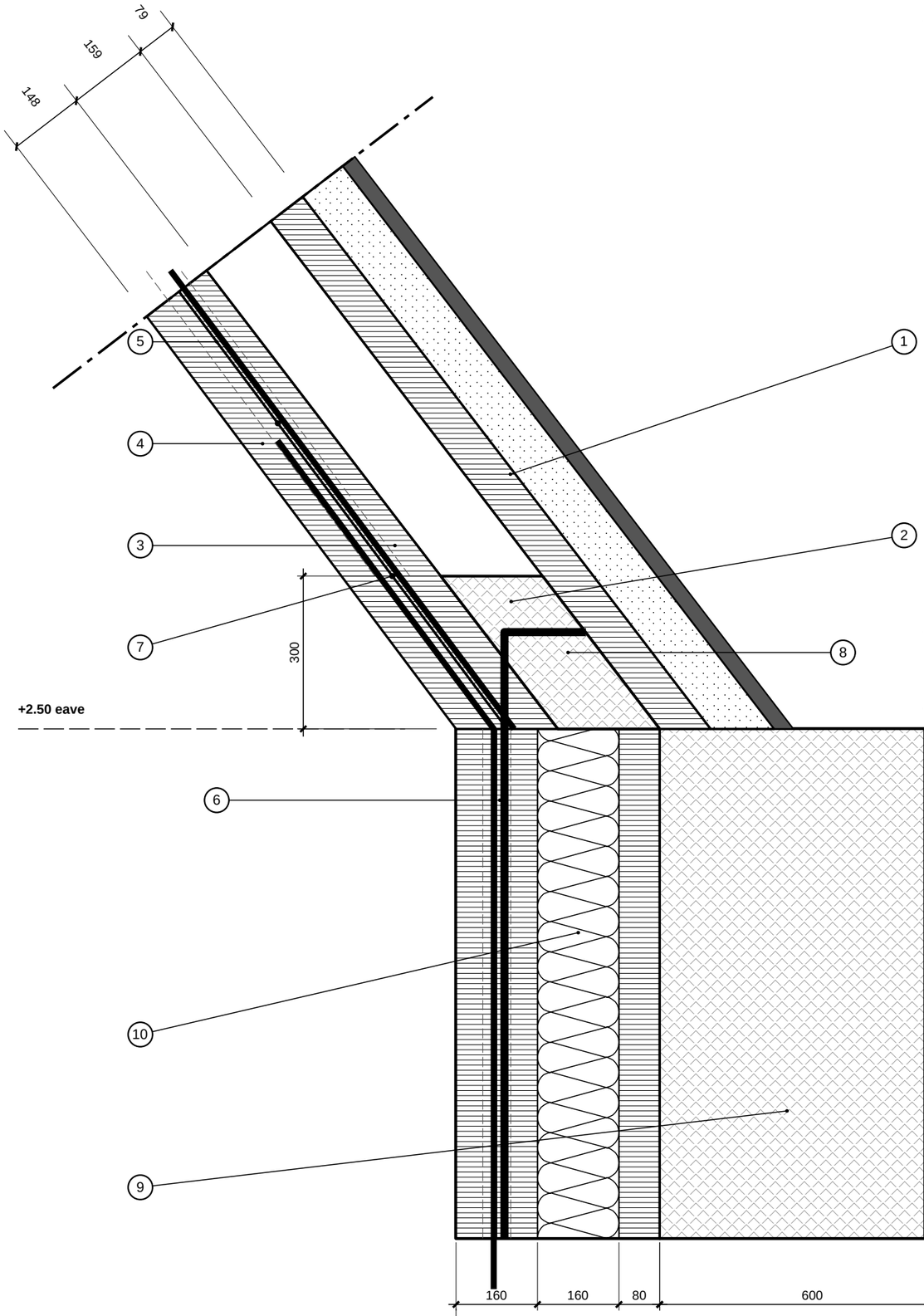
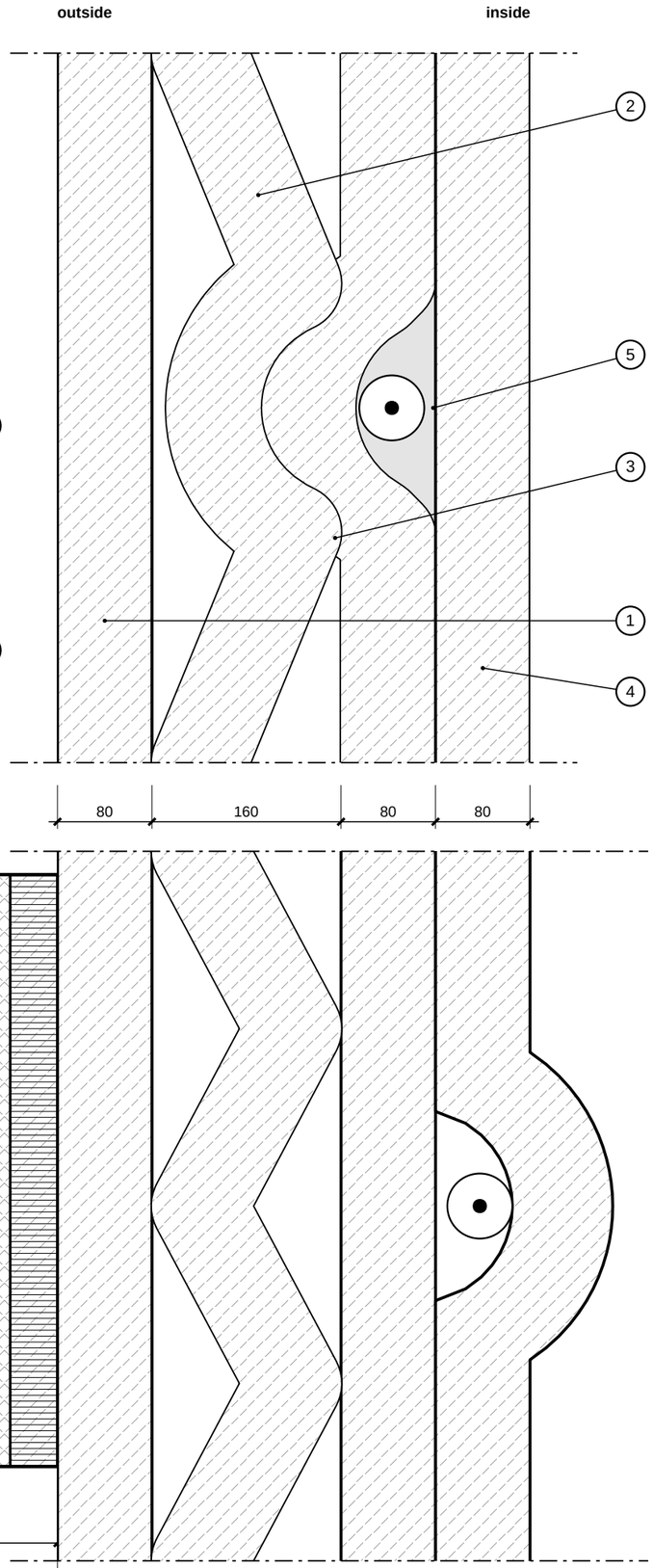


Plan of one roof piece, 1:5

layer pattern, bars every @ 600 along the path



Detail D1: eave, roof to wall

(1) Roof outer layer 80, printed with the rest of the piece; outside it the insulation 100 (dotted) and the outer layer 30 (dark).

(2) Zig-zag lattice 80 in a 160 zone; it touches the outer layer and layer 2 every @ 600; the cells between are open air pockets for ventilation.

(3) Layer 2, 80: printed bead that curves to leave a pocket for the bar.

(4) Inner layer 1, 80, room side. Layers 1 and 2 together are the structural leaf: 160 at the base, 80 at the crown (layer 2 ends).

(5*) Rebar pocket Ø55 in the leaf, grouted after placing. Roof bars R Ø12 every @ 600 along the path.

(6*) Knee bars B Ø12 every @ 600, one continuous bar over the eave: 1100 down the wall pocket (laps wall bars A), through the solid base, 650 up the roof.

(7*) Ring bars Ø12 in the pocket zone every 300, above the solid base.

(8) Solid base of the roof: over the first 300 (vertical) above the eave the 160 zone between the printed layers is filled with cast concrete instead of the open lattice; wall bars A hook into it. Replaces the ring beam.

(9) Buttress 400 wide, 1000 from the inner face (600 beyond the wall): printed formwork (80 end cap shown) filled with cast concrete; top flush with the roof base at +2.50.

(10*) Wall 400 printed, same pattern as the roof; wall bars A Ø16 run from the footing up into the solid base and end in a 160 hook (assumed).

* = assumed, to confirm

0 1 2 3 4 5 dm

Scale 1:10 at A3 · dimensions in mm · draft 1, 2026-10-11
pyramid-roof · hand estimates, not engineered